

Informatics 2.2 / Computational Intelligence

E7: Fuzzy Logic

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Computational Intelligence

“Computational Intelligence (CI) is the theory, design, application and development of **biologically and linguistically motivated computational paradigms**”
(<https://cis.ieee.org/about/what-is-ci>)

The three **pillars** of CI:

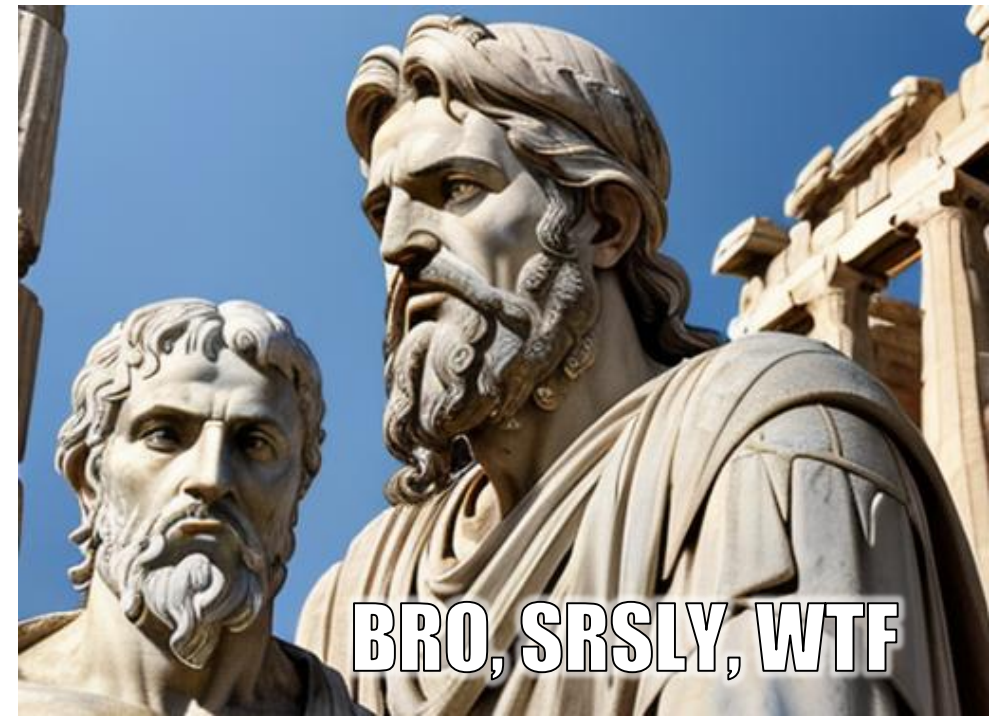
- Evolutionary Computation / Swarm Intelligence
- Neural Networks
- **Fuzzy Systems**

This lecture focuses on the **last pillar**



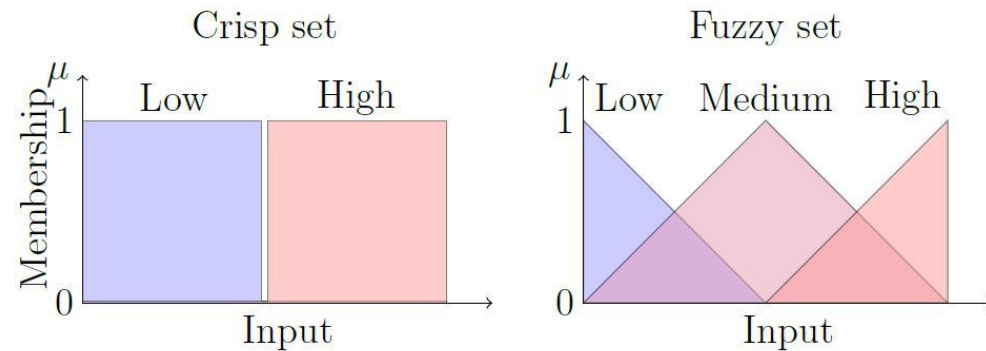
Fuzzy logic

- In Classical Boolean logic, predicates can only be **either true or false**
 - Human reasoning does not work like that
 - We express ourselves using **uncertain and vague concepts**
 - Our reasoning is robustly based on these concepts
- Simple example: the **sorites paradox**
 - Eubulides of Miletus, IV century BC:
"If a heap is reduced by a single grain at a time, at what exact point does it cease to be considered a heap?"
 - Aristoteles was not amused



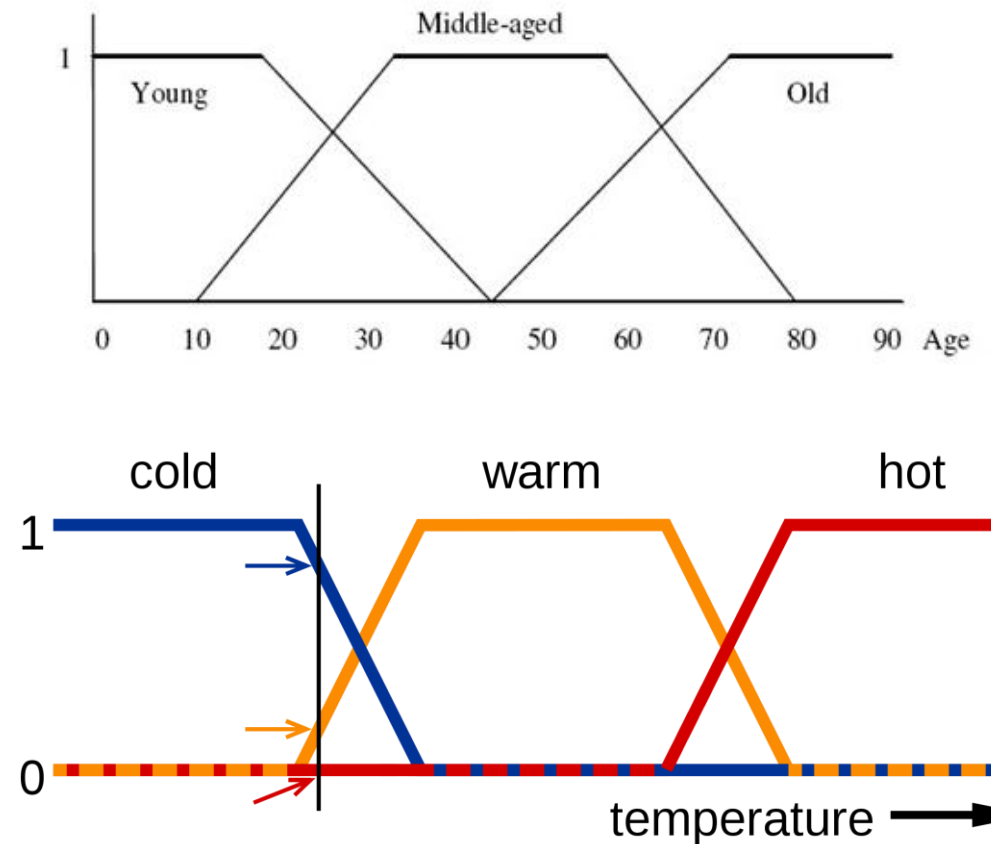
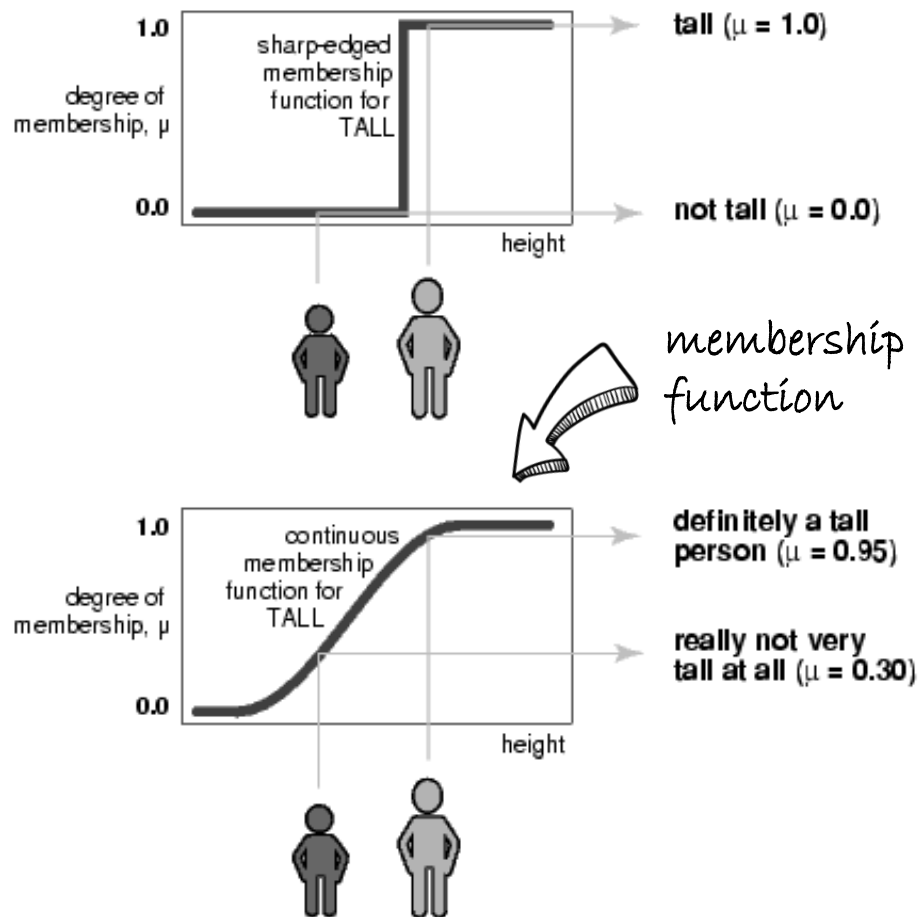
Fuzzy sets

Modeling vagueness: the notion of **fuzzy sets**

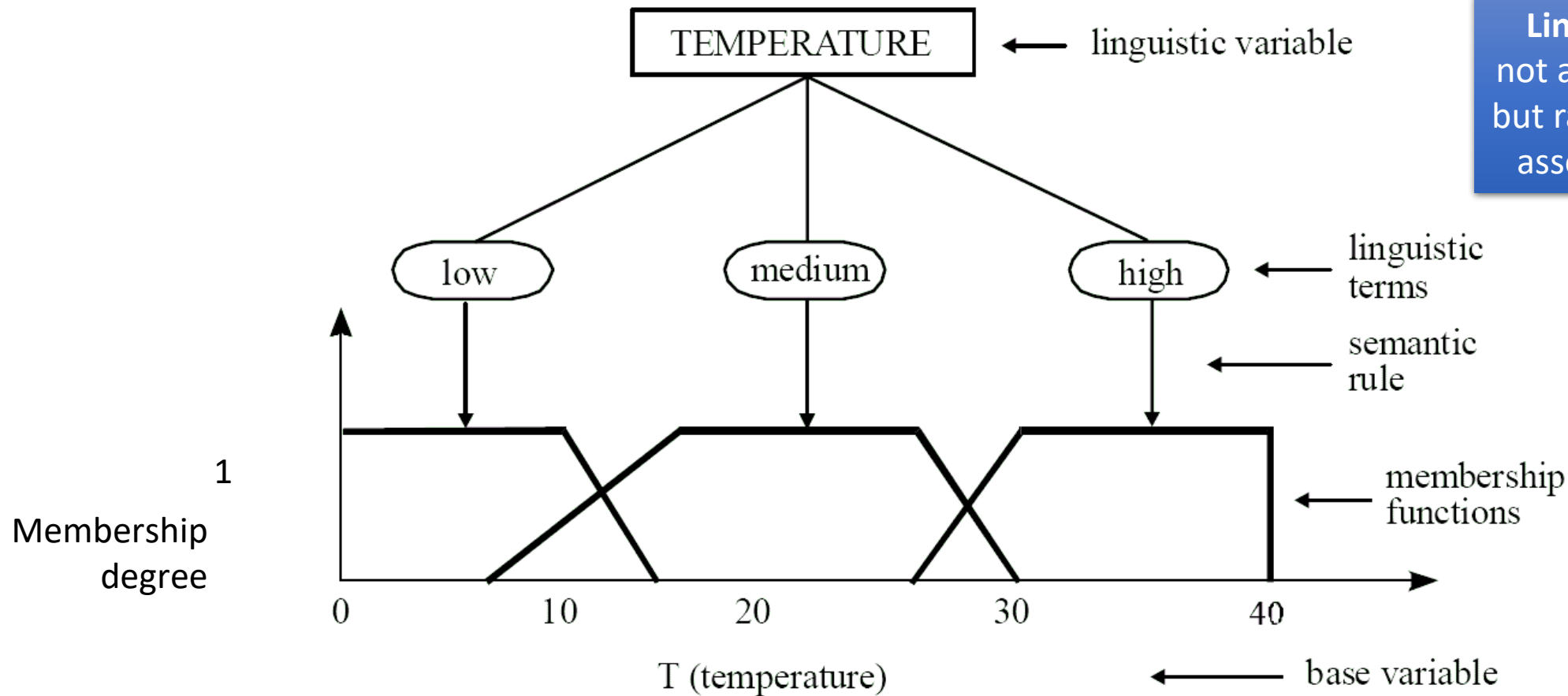


- Any element of the **Universe of Discourse belongs to some extent** to each set
- That is, the membership degree to a set ranges **from 0** (no membership at all) **to 1** (the element fully belongs to the set) instead of just 0 (false) or 1 (true)
- Fuzzy sets can describe **vagueness of concepts** and is close to **natural language**
- Connect qualitative and quantitative representations

the membership function



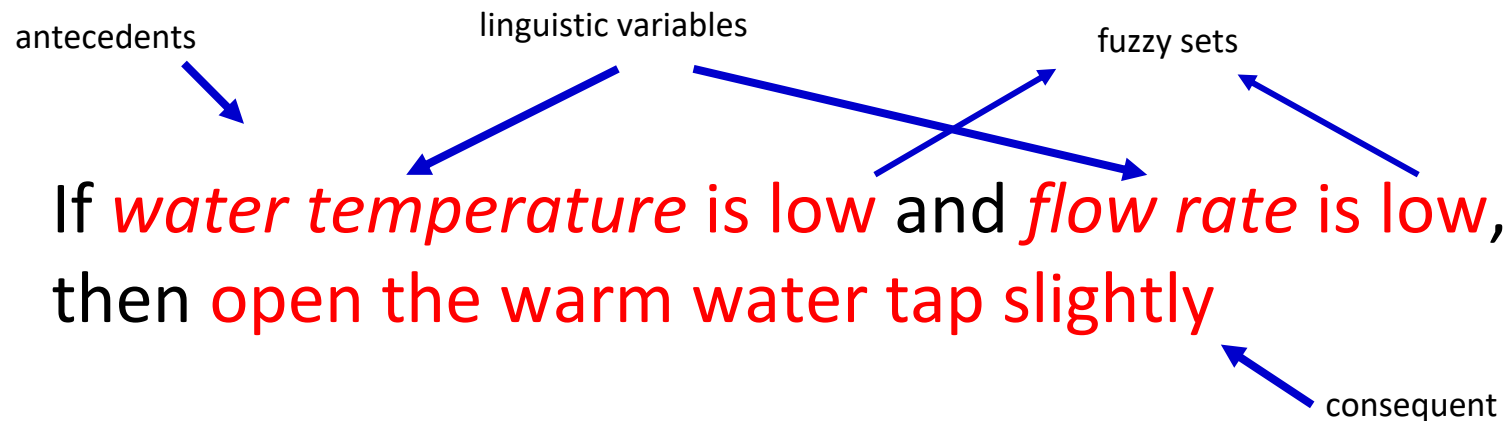
Linguistic variables and terms



Linguistic variables do not assume «crisp» values but rather **linguistic terms** associated to **fuzzy sets**

Fuzzy rules

- Fuzzy systems manipulate fuzzy sets to **model the world**
- Most fuzzy systems are based on **fuzzy rules** like:



Antecedents describe a condition in which a rule is valid locally

Consequent can be a fuzzy set, a number or another model (e.g., a linear model)
Consequents can be used to model a behavior or **drive decisions**

Output values

- The easiest way to understand outputs is to take an example:
IF *water temperature is low* AND *flow rate is low* THEN *open the warm water tap slightly*
- There can be **multiple possible outputs**, taking values that make sense in the specific application domain. For instance, in this case it could be:
 - OPEN FULLY: 100%
 - HALF OPEN: 50%
 - **OPEN SLIGHTLY : 25%**
 - CLOSE TAP: 0%
- The **higher the truth** of this antecedent ("temperature is low and flow rate is low"), the **more the output will be influenced** towards the "open slightly" value
- We can have **multiple conflicting rules** leading to all **intermediate** behaviors

Modus ponens

$$\frac{P \rightarrow Q, P}{Q}$$

Example:

- Rule: “it rains $\rightarrow$ the ground is wet”
- Fact P : “it is raining”
- Consequent Q : “the ground is wet”

Generalized modus ponens

$$\frac{P \rightarrow Q, P'}{Q'}$$

Example:

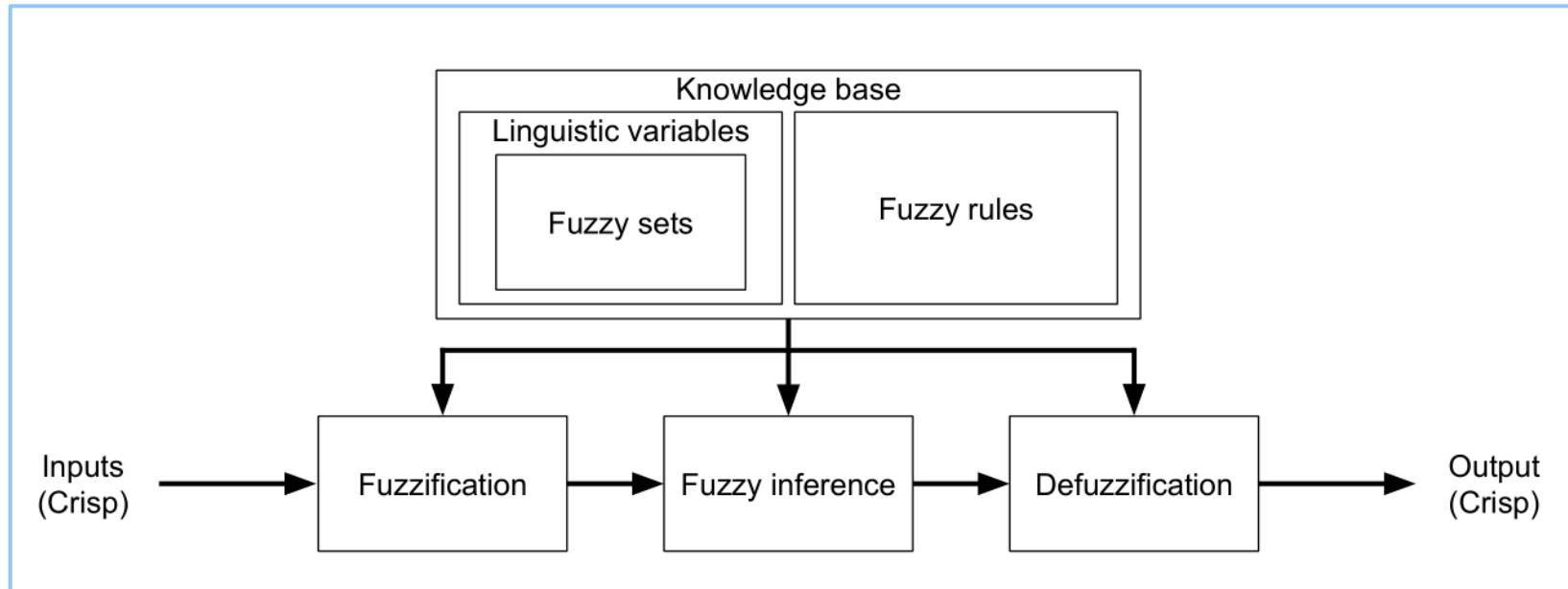
- Rule: “tall people are heavy”
- Fact P' : “John is quite tall”
- Consequent Q' : “John is rather heavy”

Fuzzy “if-then” rules

- General format of fuzzy rules: **if x is A then y is B**
- Examples of fuzzy rules:
 - *if pressure is high, then volume is small*
 - *if the speed is high, then apply the brake a little*
 - *if demand is rising fast and quantity-on-hand is low, then order amount is positive large*
 - *if spaghetti is very soft then pasta is scotta*
 - *if many cells look like a tumor then patient has cancer*
- The rules are logical statements and can have **arbitrary complexity**

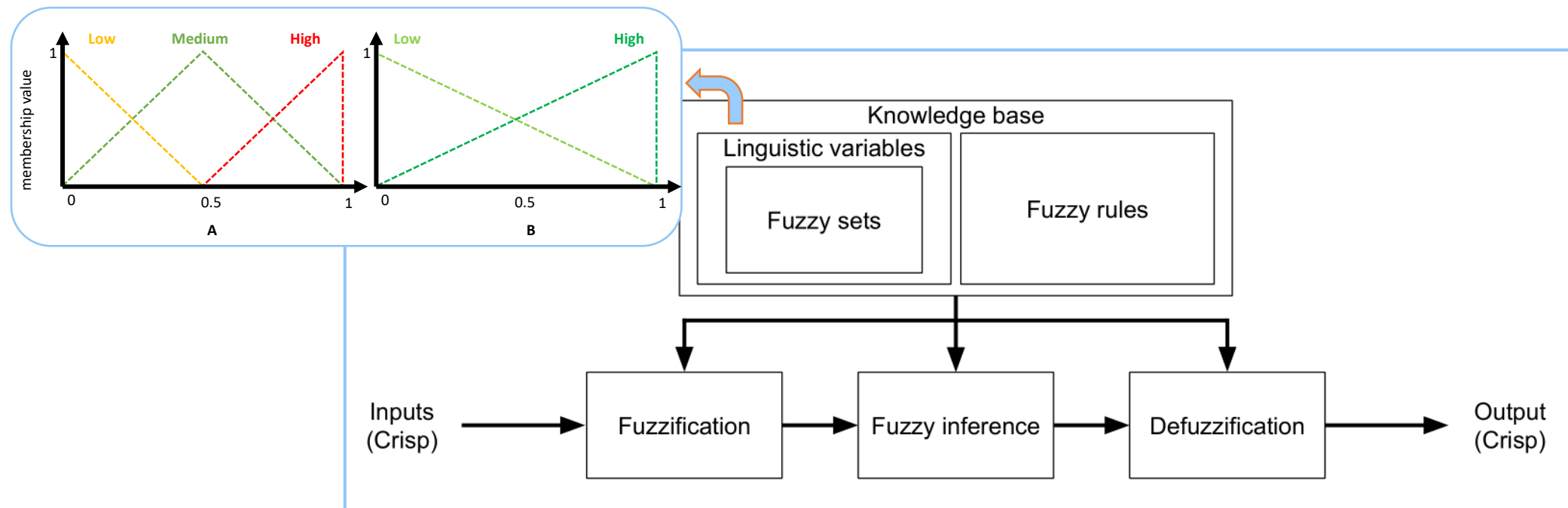
Fuzzy inference systems

- FIS exploit a knowledge base composed of a **set of linguistic variables** and **fuzzy rules**
- Fuzzy inference **maps a given input to an output** by means of fuzzy logic



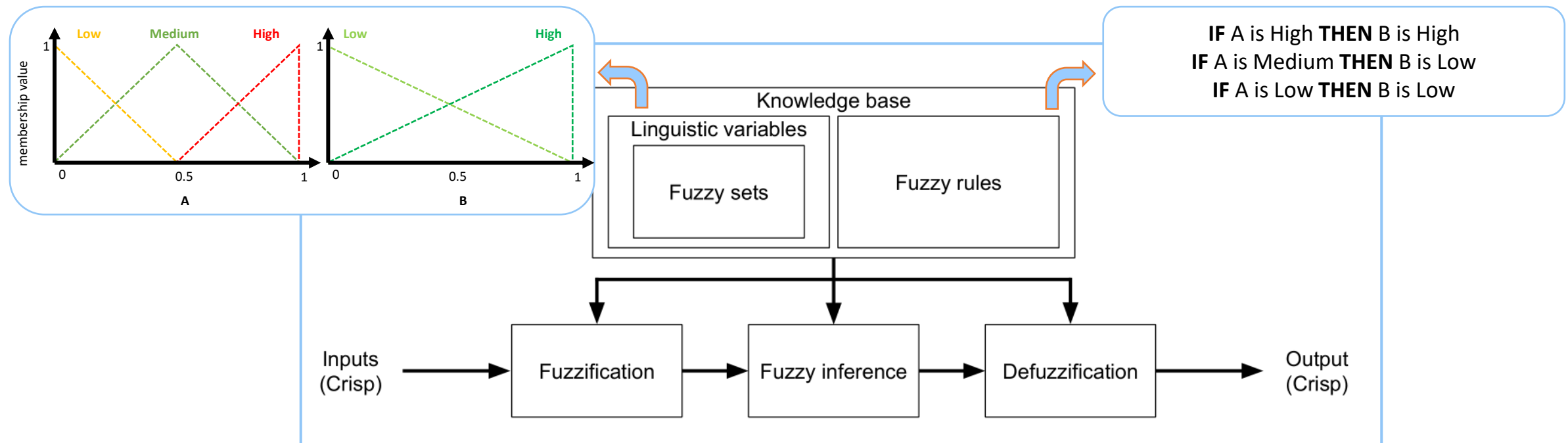
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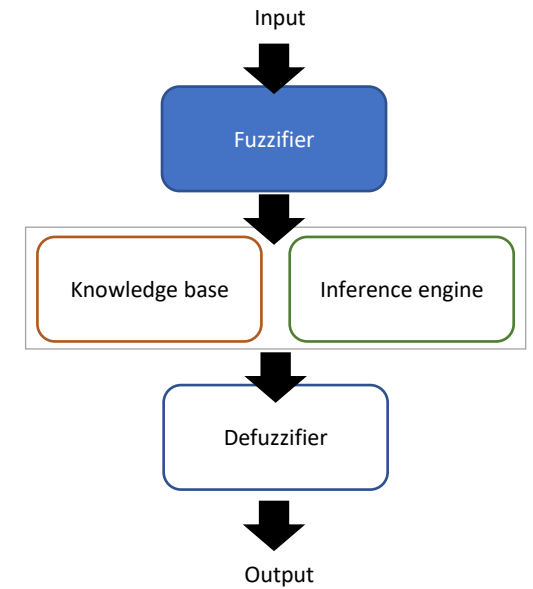
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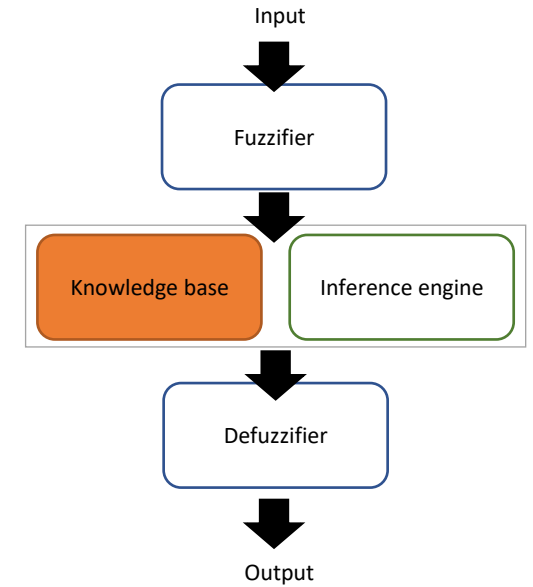
Fuzzifier

- Represents the **interface** between the “outside” world and the fuzzy system (input side)
- Determines the **match** between a given **input** and the **linguistic terms**
- Given a **crisp** input it is responsible for the computation of all **memberships values** of linguistic terms
- An actual measure of the outside world gets “fuzzified”



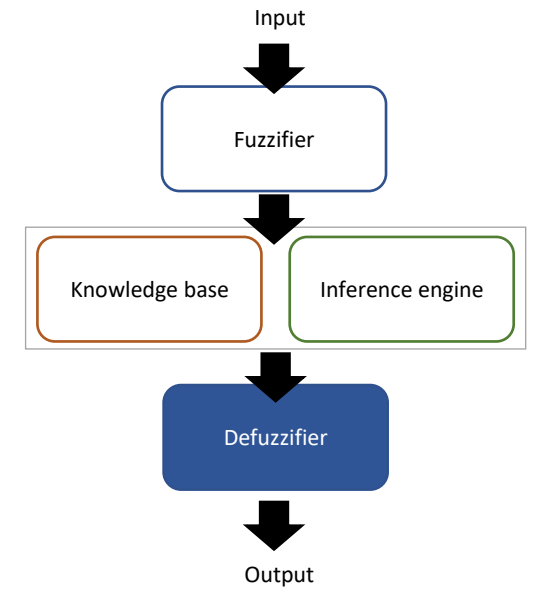
Knowledge (or rule) base

- Heart of the knowledge base of the fuzzy system
- Encodes the **relationships between inputs and the outputs**
- Rules can be **examples, rules of thumb**, encoded **experience, qualitative** relations between variables, etc.
- Rules are generally represented as **if-then statements**



Defuzzifier

- **Interface** between the **FIS** and the “**outside**” world (output side)
- **Needed when a crisp output is required** (e.g. a final decision, a control action, a final advice)
- **Computes a number/symbol** that represents the **output fuzzy set**
- **Several** defuzzification methods for **output fuzzy sets** exist (more about this later)



Aggregation of rules and logic operators

Antecedents can be combined in a rule by using logic operators:

- AND = $\min()$
- OR = $\max()$
- NOT = $1 - \mu()$
- We can have **multiple rules** whose antecedents are satisfied to a certain degree
- The output of multiple rules, affecting the same consequent, must be aggregated...

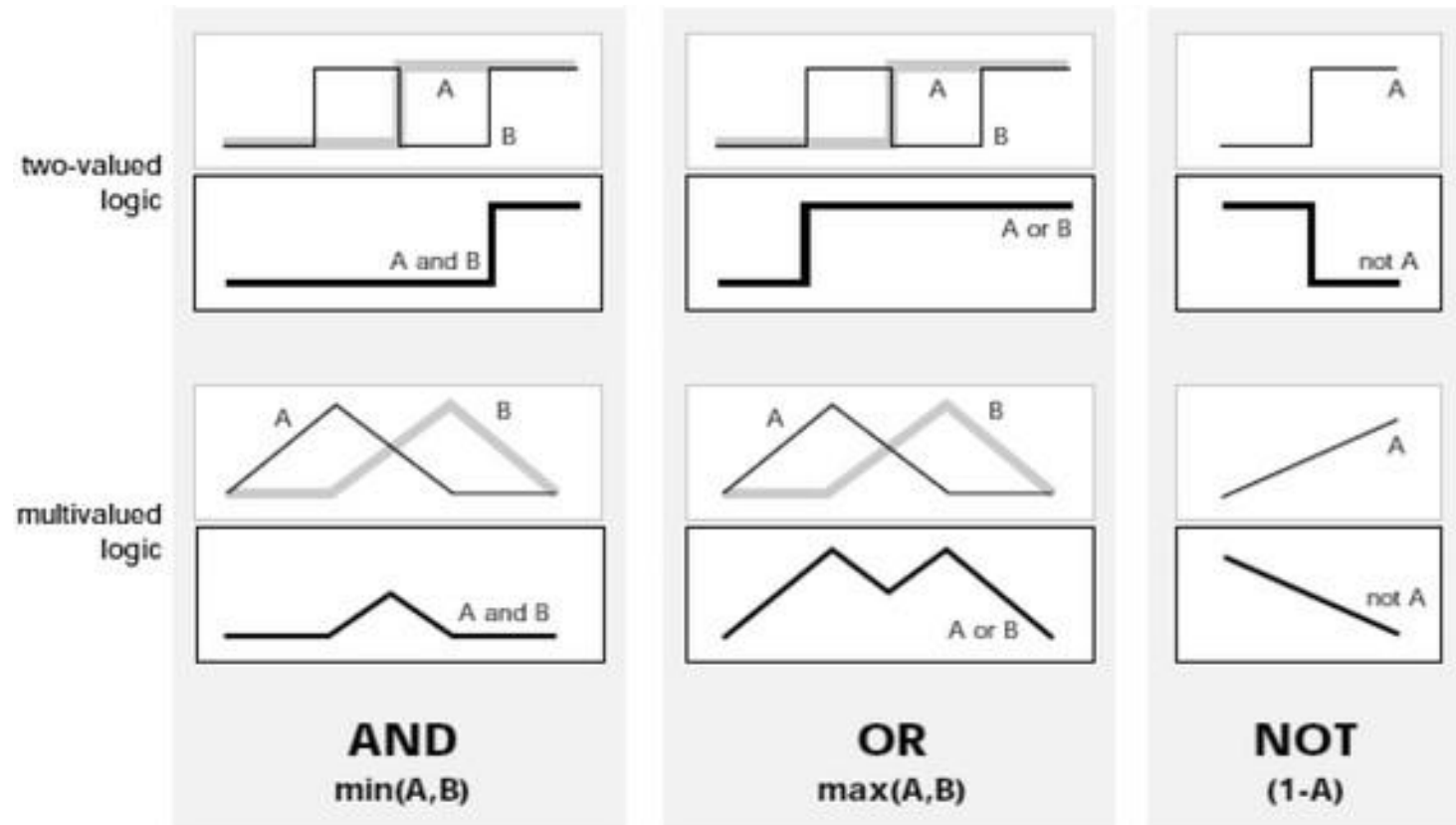
IF A is High **AND** B is Low **THEN** C is High

IF A is High **OR** B is Low **THEN** C is High

IF A is **NOT** High **THEN** B is High

IF A is High **THEN** B is High
IF A is Medium **THEN** B is Low
IF A is Low **THEN** B is Low

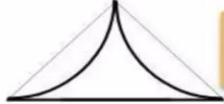
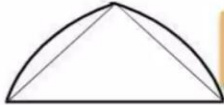
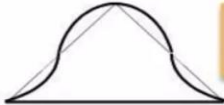
Fuzzy logic operators



Hedges

- **Linguistic operators** implementing adverbs like very, somewhat, slightly, etc.

Representation of hedges in fuzzy logic

<i>Hedge</i>	<i>Mathematical Expression</i>	<i>Graphical Representation</i>
Very very	$[\mu_A(x)]^4$	 Concentration
More or less	$\sqrt{\mu_A(x)}$	 dilation
Somewhat	$\sqrt{\mu_A(x)}$	 dilation
Indeed	$\begin{aligned} &2 [\mu_A(x)]^2 && \text{if } 0 \leq \mu_A \leq 0.5 \\ &1 - 2 [1 - \mu_A(x)]^2 && \text{if } 0.5 < \mu_A \leq 1 \end{aligned}$	 intensification

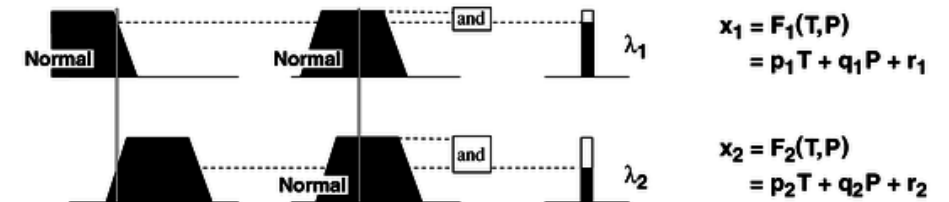
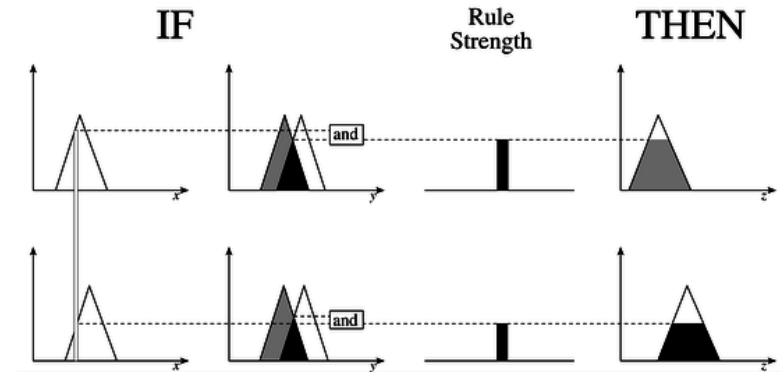
Fuzzy Inference

- The **degree of fulfilment** of a rule determines **how much that rule is valid**, and that ultimately depends on the **degree of satisfaction of the rule's antecedents** ("the more antecedents are true $\rightarrow$ the higher the fulfilment")
- Fuzzy Inference computes the **output of each fuzzy rule** in the rule base, **given the degree of fulfilment** of that rule
- Two most important Fuzzy Inference methods: **Mamdani** and **Sugeno**

Interpretation of a fuzzy rule

The two **most used** (and widely accepted) inference methods are:

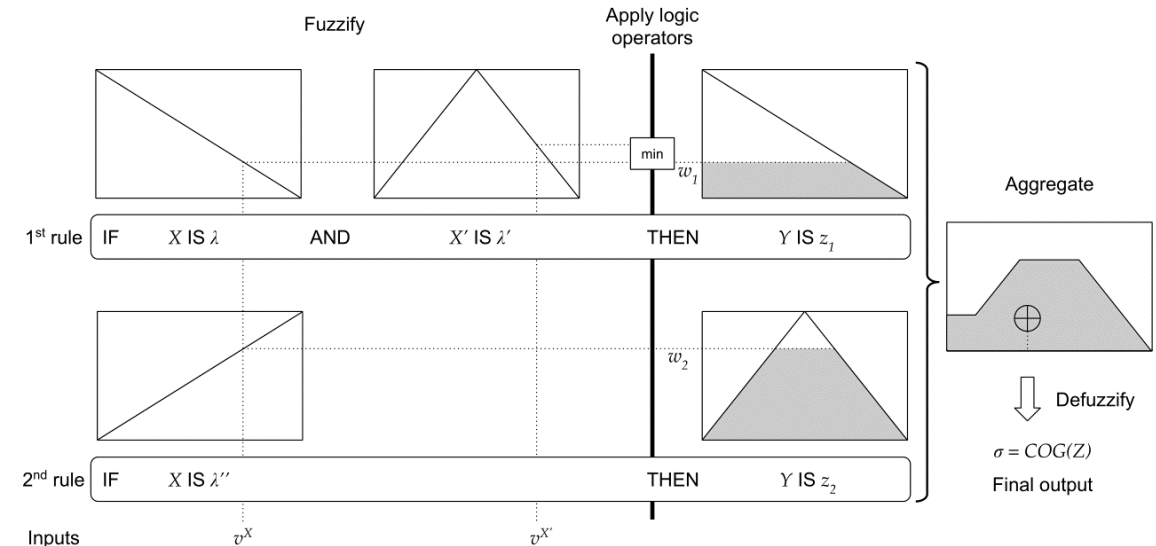
- **Mamdani method**: the **consequent** of a rule is a **fuzzy set** [Mamdani *et al.* 1975]
- **Sugeno**: the **consequent** of a rule is a **function** [Sugeno *et al.* 1985]



Mamdani fuzzy inference

Mamdani method

- **Consequents are fuzzy sets**
- Fuzzification of inputs
- Application of fuzzy operations
- Implication
- Aggregation of outputs
- **Defuzzification step** (next slide)



Historically, this is the first proposed method [[Mamdani and Assilian, Int J Man-Machine Studies 1975](#)]

- **Easy to understand** and easy to **formulate rules**
- Usually applied for **knowledge-driven** approaches
- Strongly dependent on aggregation/defuzzification
- **Many firing rules can give unreliable results**

Defuzzification Methods

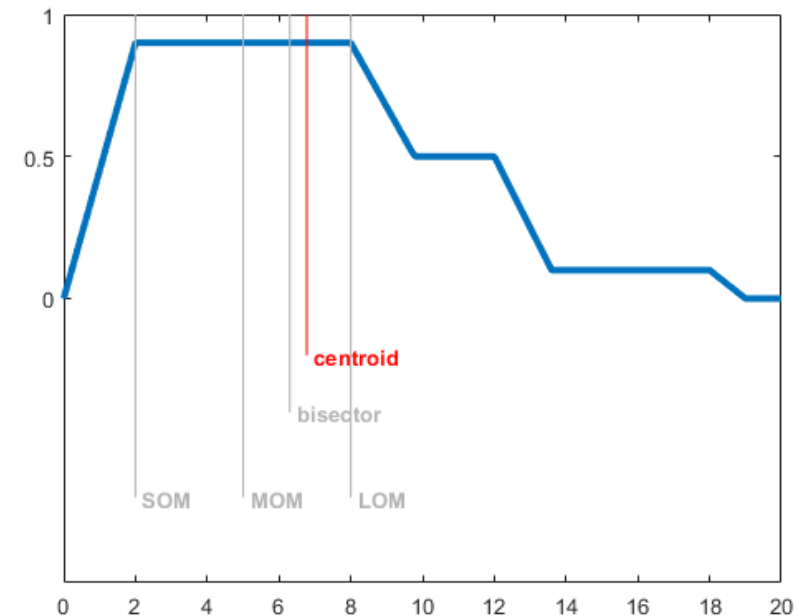
- Center of Gravity (aka centroid)
$$z^* = \frac{\int_Z \mu_A(z) z dz}{\int_Z \mu_A(z) dz}$$

- Bisector of area
$$\int_{-\infty}^{z^*} \mu_A(z) dz = \int_{z^*}^{\infty} \mu_A(z) dz$$

- Mean of maximum (aka MoM)
$$z^* = \frac{\int_{Z'} z dz}{\int_{Z'} dz}, \quad Z' = \{z \mid \mu_A(z) = \mu^*\}$$

- Smallest of maximum (aka SoM)
$$\min_{z \in Z'} z$$

- Largest of maximum (aka LoM)
$$\max_{z \in Z'} z$$



These methods return different defuzzified values z^* for the same fuzzy set

Example of Mamdani reasoning

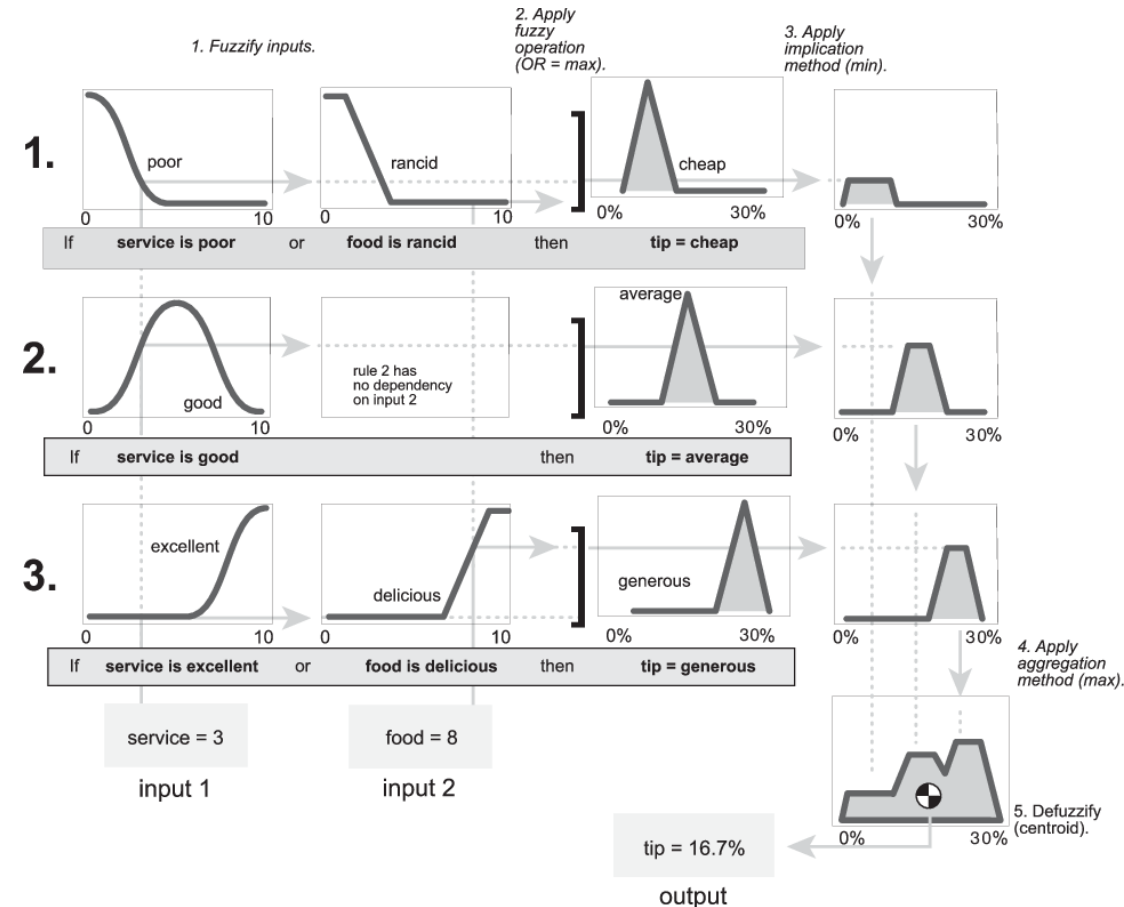
Tipping example, 3 rules:

1. If service is poor or food is rancid then tip is cheap
2. If service is good then tip is average
3. If service is excellent or food is delicious then tip is generous

The antecedents are **connected with OR**:
we will take the **maximum** of the fuzzy inputs

Mamdani system: **consequents are fuzzy sets**

- The implication method is the **minimum of the consequent** (see how the fuzzy set is “cut”)
- **Responses are aggregated** with maximum
- **Centroid** (=center of gravity) defuzzification is used and the **final tip of 16.7%** is returned as output



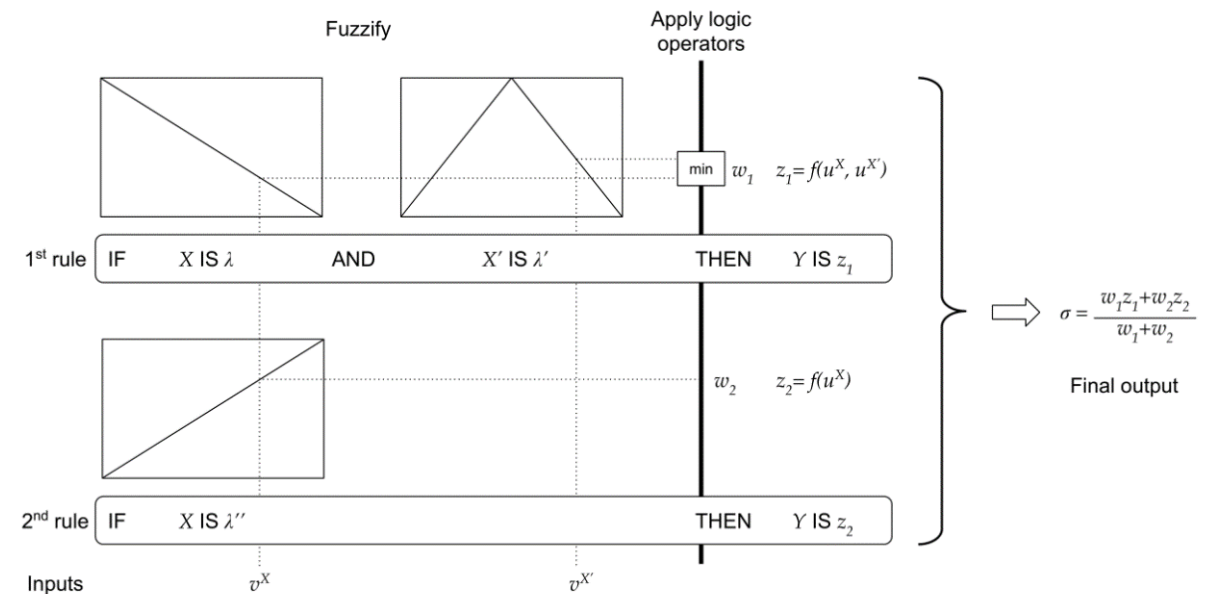
Sugeno fuzzy inference

The Sugeno (aka Takagi-Sugeno-Kang) method is one of the most used inference approaches

- **Consequents are functions**
- Fuzzification of inputs
- Application of fuzzy operations
- Implication
- Aggregation of outputs (weighted average)

Consequent is a **function** (or a **scalar** in the case of 0-order Sugeno systems)

- **No need to defuzzify** the outputs
- Computationally **efficient**
- **Guaranteed continuity** of the output surface
- **Higher order systems** (e.g., complex output functions) affect interpretability (but would improve accuracy to the level of deep neural networks!)



Sugeno fuzzy models and reasoning

- **Zero-order Sugeno: constant** consequent
if x is A and y is B then $z = c$
- **First-order Sugeno: linear** consequent
if x is A and y is B then $z = ax+by+c$
where a, b, c are scalar values
- And so forth...
- **Sugeno reasoning:** output is a **weighted average** of individual rule outputs

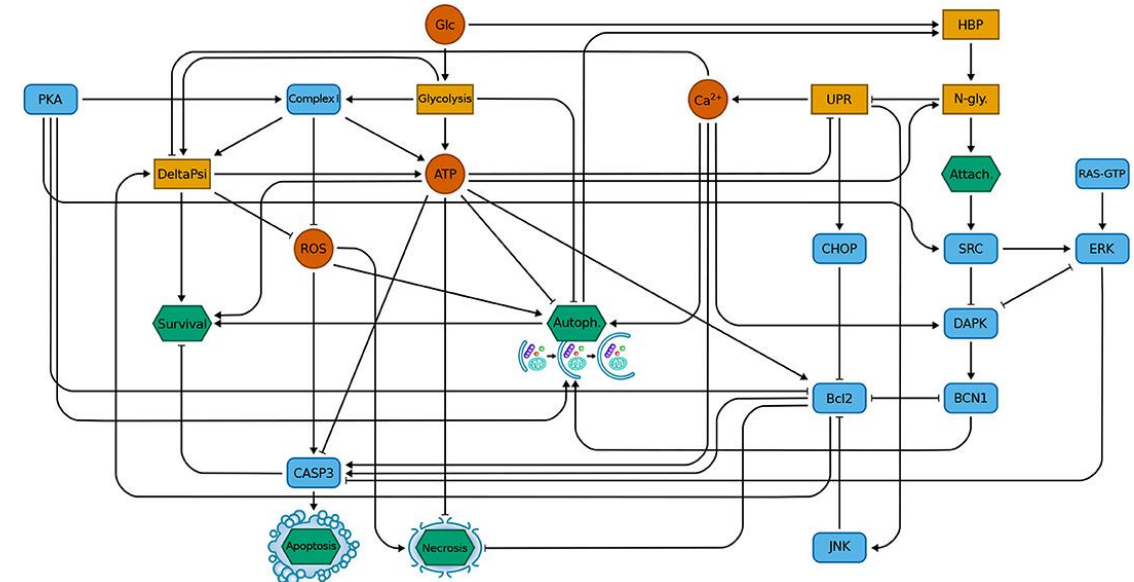
$$z^* = \frac{\sum_{i=1}^N \beta_i z_i}{\sum_{i=1}^N \beta_i}$$

Consequent (output) of rule i

Degree of fulfillment of rule i
(same as with Mamdani)

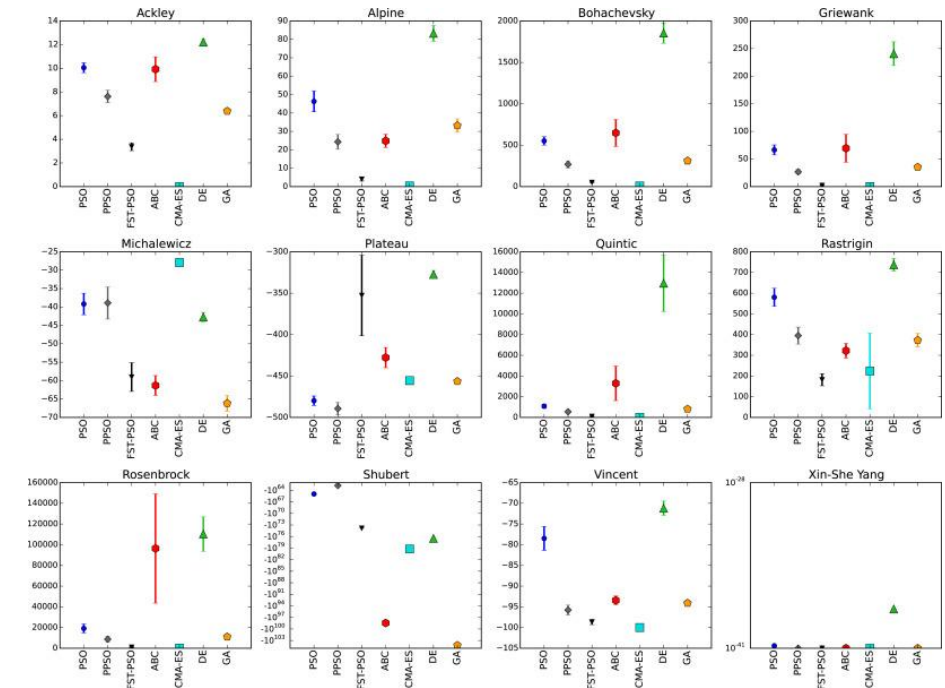
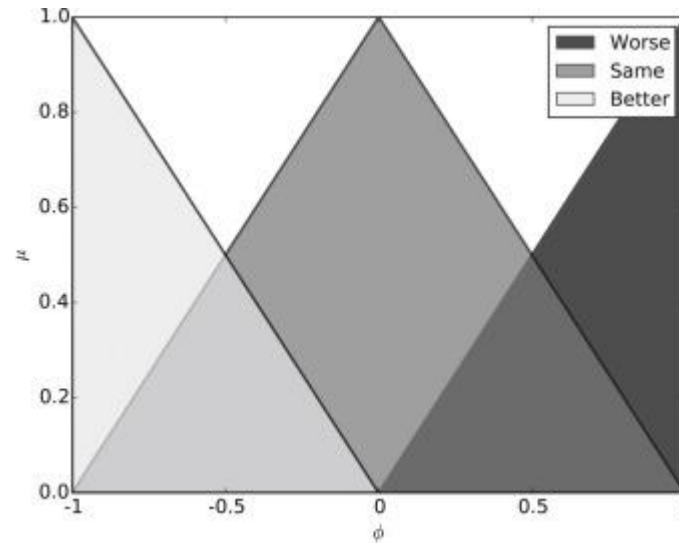
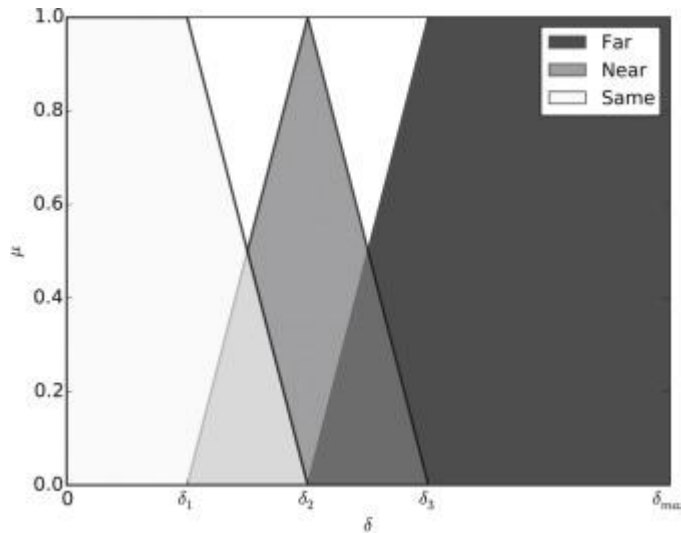
Example: dynamic fuzzy modeling

- We created a FIS to create a **phenomenological dynamic model of the functioning of cancer cells**
 - **>200 rules**, each based on a specific scientific paper
 - Example of rule: «IF glucose is high THEN Glycolysis is high»
 - **Network** of interacting linguistic variables: **complex and unpredictable dynamics**
 - **0-order Sugeno inference**
 - Several experiments for **validation**
 - [Nobile et al., Bioinformatics 2020],
[Nobile et al., Front Genet 2021]
- NSGA-II algorithm to **identify optimal drug combinations**: you will see that in the second part of the course

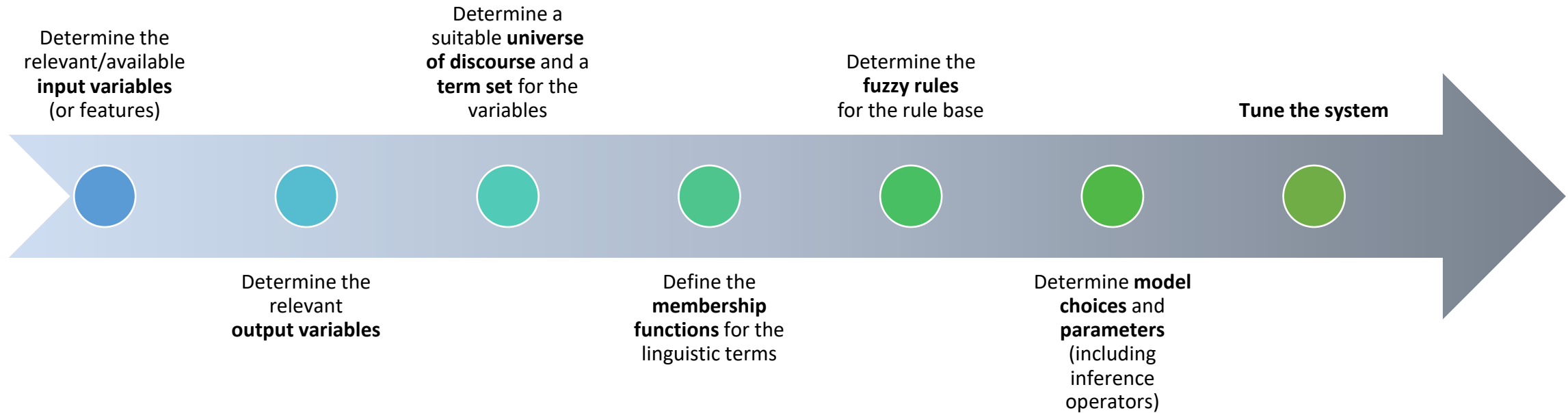


Example: FST-PSO

- We used a FIS to implement a **self-tuning variant of the PSO optimization algorithm**
 - FST-PSO uses **rules** like *"when a particle is near to the best particle, reduce the cognitive factor"* or *"when the fitness is better than the previous iteration increase the inertia"*
 - **Outperforms classic PSO**, competitive with state-of-the-art, you will see that in the next lectures
 - [Nobile et al., Swarm Evol Comput 2018]



Creating a Fuzzy System



Data-driven fuzzy modeling

- We can “**learn**” **FIS from data!**
- Automatically build a FIS whose response/answer **fits with available data**
- That is, using a **Machine Learning** approach
- The AI would be based on **fuzzy reasoning** instead of artificial neurons
- We aim at producing a **human-readable** model that **minimizes the error** with respect to the data-set

Why Fuzzy Inference Systems?

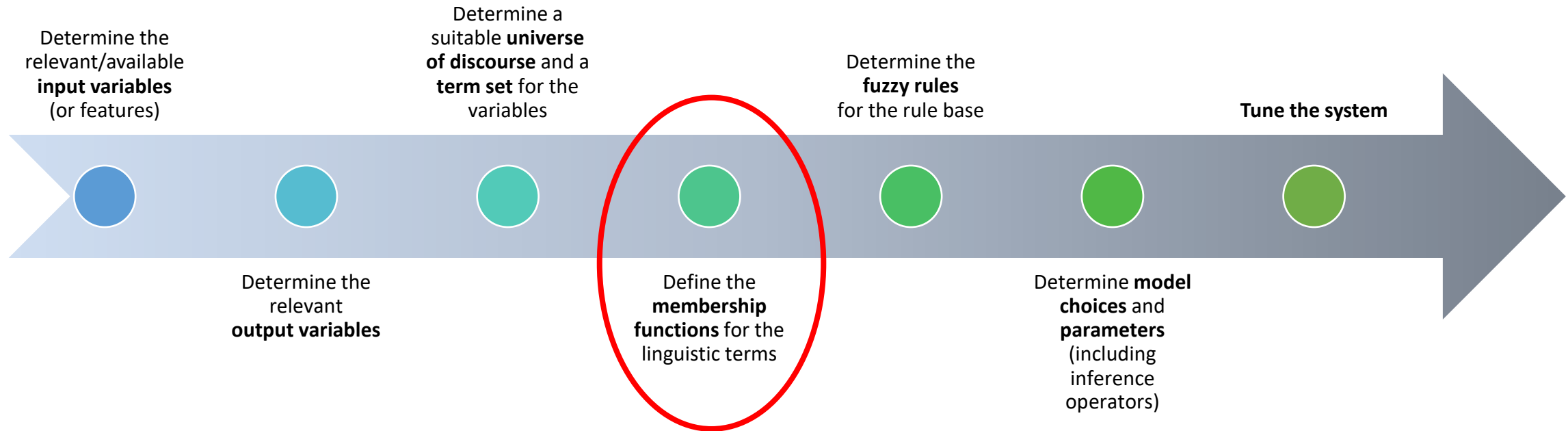
FIS are **universal approximators**: it was proven that **a sufficiently complex FIS can model any function**

Why using FIS instead of, say, NNs? → **Interpretability!**

- FIS are close to natural language: they **exploit linguistic variables** and **human-comprehensible rules** built using logic constructs
- **Rules can be inspected, changed, removed** → generate trust
- **Additional rules** can be added to embed **domain expertise** (this is basically impossible with NNs)
- **Interpretable decisions can now be trusted** (e.g., healthcare)
- Fuzzy sets represent bridges between symbolic processing and numerical computations



Creating a Fuzzy System



Clustering

A cluster is a **group of objects** that are more **similar** to one another than members of other clusters

- In metric spaces, similarity is often defined by means of a **distance** norm
- **Different measures of similarity** and different clusters will be appropriate for different purposes
- Similarity is a fuzzy concept, after all
- Can we automatically identify such groups in data?

What data are we talking about?

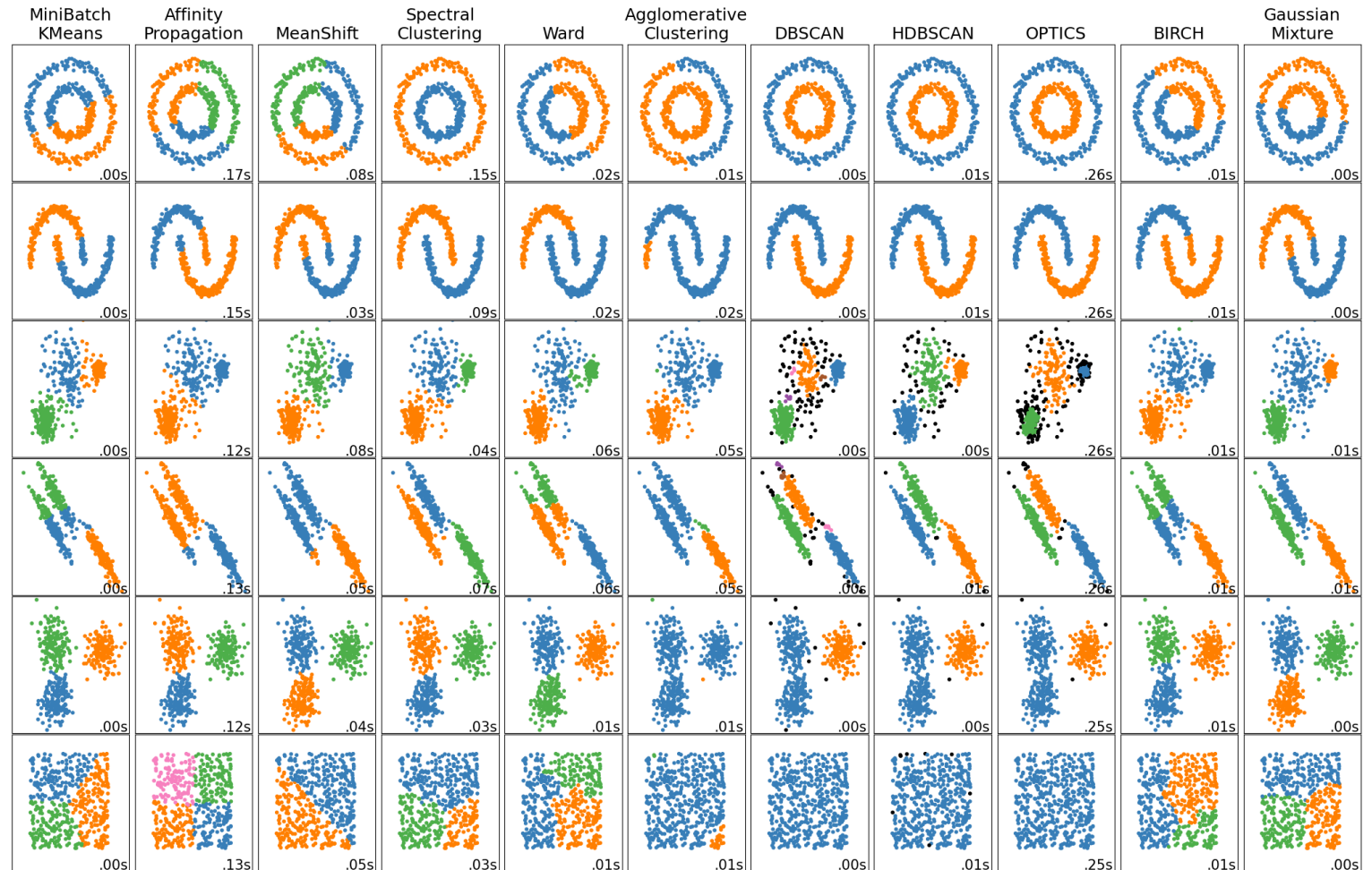
- Assume that we have **N samples** of some experimental measurements (e.g., objects, patterns, pixels, cells, patients, ...)
- Each sample is characterized by **D features** or **attributes**
- We denote by $\mathbf{x}_k \in \mathbb{R}^D$ the k -th sample, where $k = 1, \dots, N$
- The collection of all data is denoted by $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_N)$

Clusters [1/2]

Look at this figure (from SciKit-learn documentation)

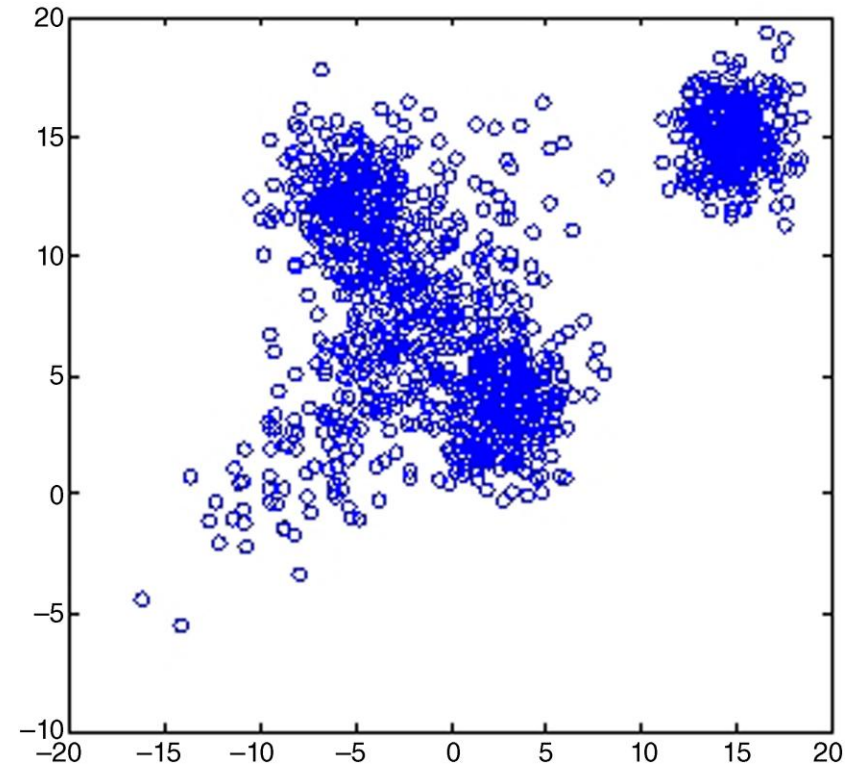
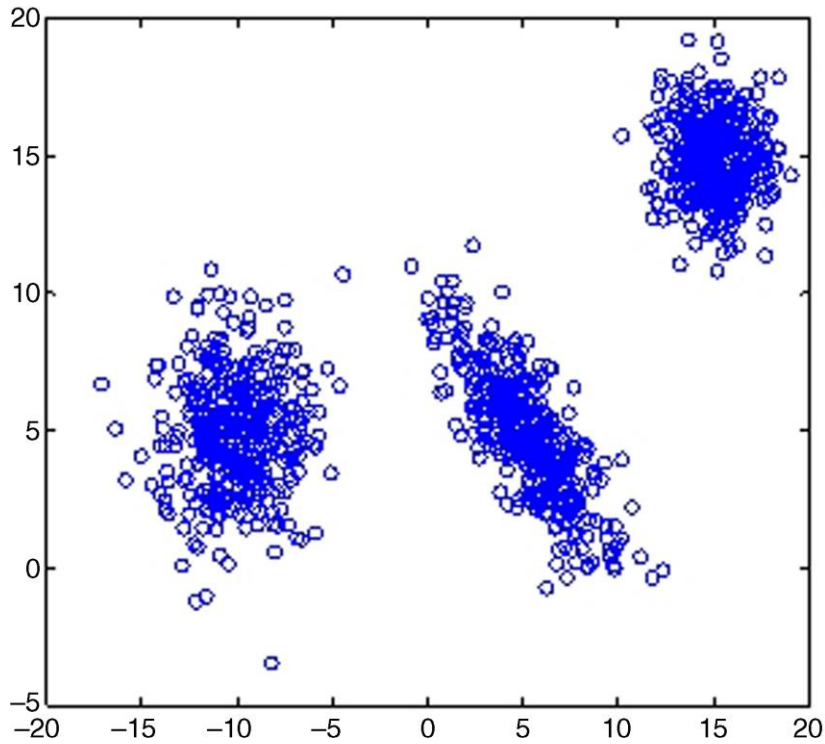
Data can reveal **groups** of different geometrical shapes, sizes and densities

Some algorithms are better than others at detecting **specific structures** in data



Clusters [2/2]

- How many clusters do you see in the following figures?



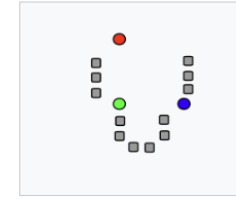
K-means

- Algorithm for **partitioning a set of samples** into **K different groups** (=clusters)
 - **NP-hard** problem!
 - **Unsupervised** (no labels nor values to predict)
- Specifically, K-means looks for the **optimal positioning** of the **K cluster centers**, such that each cluster center represents a «prototype» of a cluster
 - Note that the prototype might be in a position never observed
 - The algorithm looks for the optimal solution with respect to minimization of **intra-cluster variance** (rationale: the samples in a cluster should be **as similar as possible**)

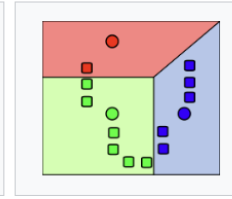
K-means functioning

- Classic approach: **Lloyd algorithm**
 - Slow, not efficient
 - Very dependent on initialization
 - **Converges to local minimum**
- Alternative: use **PSO!**
 - Even slower!
 - **Not trapped** in local minima
 - See [\[Spolaor et al., FUZZ-IEEE 2019\]](#)

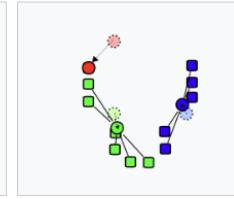
Demonstration of the standard algorithm



1. k initial "means" (in this case $k=3$) are randomly generated within the data domain (shown in color).



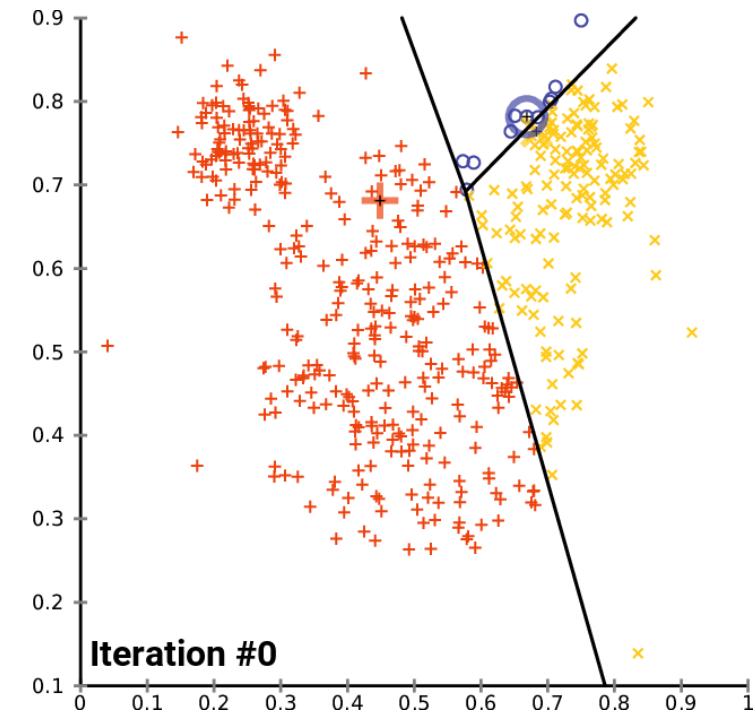
2. k clusters are created by associating every observation with the nearest mean. The partitions here represent the [Voronoi diagram](#) generated by the means.



3. The [centroid](#) of each of the k clusters becomes the new mean.



4. Steps 2 and 3 are repeated until convergence has been reached.



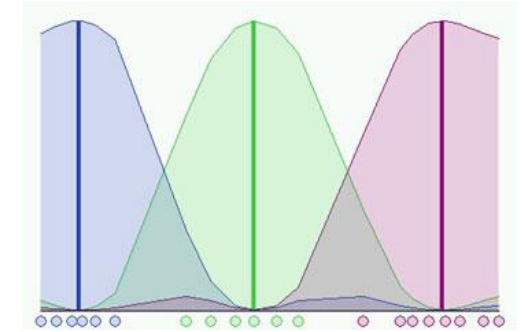
Types of clustering

- Type of **subsets**
 - Hard clustering
 - **Fuzzy clustering**
- **Algorithmic approach**
 - Agglomerative hierarchical methods
 - Splitting hierarchical methods
 - **Objective function**
- A huge number of (hard or fuzzy) clustering algorithms exist
- We will focus on the most widespread: **Fuzzy C-means**

Types of subsets (=crisp vs fuzzy clustering)

"Crisp" clustering algorithms

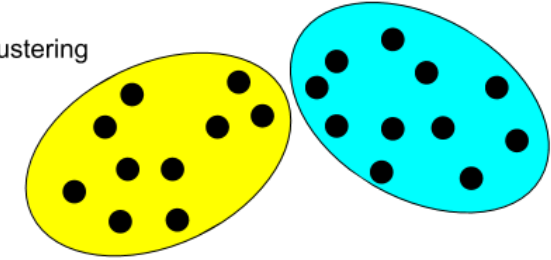
- Partition the data set into disjoint groups, i.e., each datum belongs to one cluster only
- Similarity is quantified using some metric
- E.g., k-means



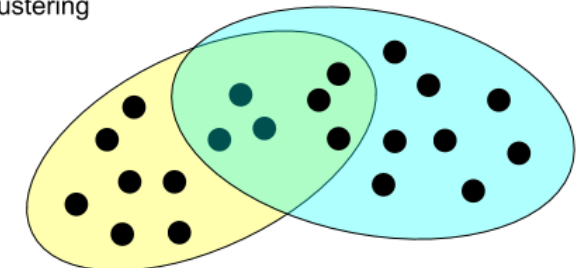
Fuzzy clustering algorithms

- Partition the data set into **overlapping groups**, i.e., each datum can belong to multiple clusters with **varying degree of membership**
- Similarity is quantified using some metric which is modified by **membership values**
- E.g., **fuzzy C-means (FCM)**

Crisp clustering



Fuzzy clustering



Fuzzy C-means [1/2]

Objective function-based algorithm to partition data into **fuzzy groups**

- Partitioning according to **similarity**
- One of the most popular clustering algorithms

- Given the data $\mathbf{X}$

$$\mathbf{U} = \begin{bmatrix} \mu_{11} & \dots & \mu_{1N} \\ \vdots & \ddots & \vdots \\ \mu_{C1} & \dots & \mu_{CN} \end{bmatrix}, \quad \mu_{ij} \in [0,1]$$

- Find the partition matrix

- And the cluster centers

$$\mathbf{V} = \{\mathbf{v}_1, \dots, \mathbf{v}_C\}, \quad \mathbf{v}_i \in \mathbb{R}^n$$

Fuzzy C-means [1/2]

Objective function-based algorithm to partition data into **fuzzy groups**

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- Find the partition matrix

← This matrix partitions the
N objects into
C overlapping groups

- And the cluster centers

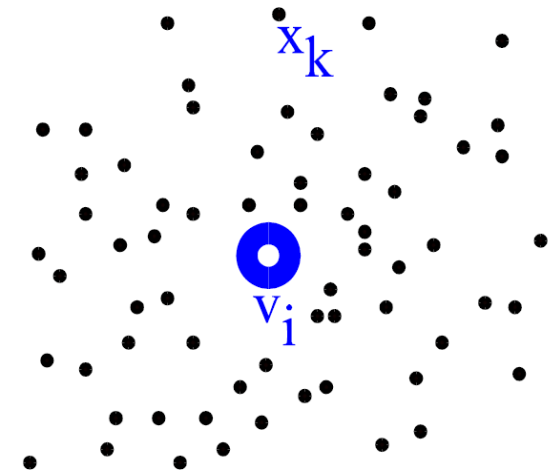
$$\mathbf{V} = \{\mathbf{v}_1, \dots, \mathbf{v}_C\}, \quad \mathbf{v}_i \in \mathbb{R}^n$$

Fuzzy C-means [2/2]

FCM minimizes the objective function: $f(\mathbf{X}, \mathbf{U}, \mathbf{V}) = \sum_{i=1}^C \sum_{j=1}^N \mu_{i,j}^{\phi} d(\mathbf{x}_j - \mathbf{v}_i)$

Subject to the following **3 constraints**:

- **Membership degree must be valid:** $0 \leq \mu_{i,j} \leq 1, \quad i = 1, \dots, C, \quad k = 1, \dots, N$
- **Memberships must sum to 1:** $\sum_{i=1}^C \mu_{i,k} = 1, \quad k = 1, \dots, N$
- **No empty clusters:** $0 < \sum_{k=1}^N \mu_{i,k} < N, \quad i = 1, \dots, C$
- $\phi \in (1, \infty)$ is the **fuzziness parameter** (usually $\phi = 2$)



The fuzzy C-means algorithm

- initialize $\mathbf{U}$
- repeat
 - compute cluster centers: $\mathbf{v}_i = \frac{\sum_{k=1}^N \mu_{i,k}^\phi \mathbf{x}_k}{\sum_{k=1}^N \mu_{i,k}^\phi}$ for all $i = 1, \dots, C$
 - update the partition matrix $\mathbf{U}$: $\mu_{i,k} = \frac{1}{\sum_{j=1}^C \left(\frac{\|\mathbf{x}_k - \mathbf{v}_i\|}{\|\mathbf{x}_k - \mathbf{v}_j\|} \right)^{\frac{2}{\phi-1}}}$
 - until termination criterion (e.g., $\|\Delta \mathbf{U}\| < \varepsilon$)

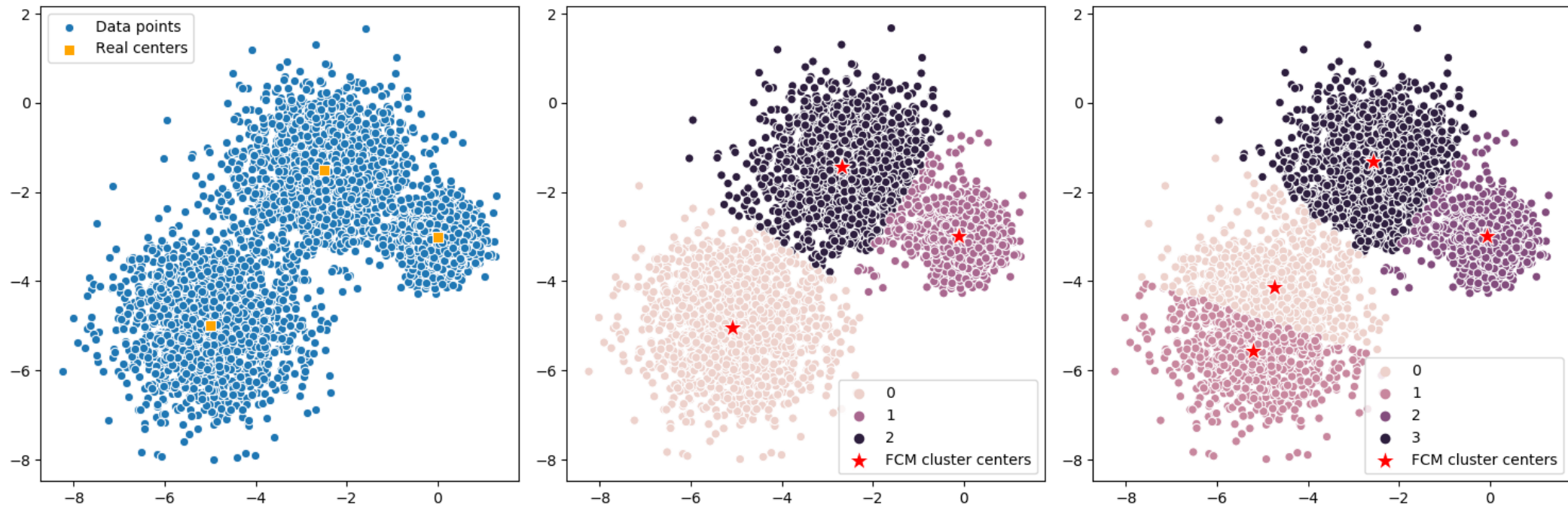
Assumes that partition matrix is fixed

Assumes that cluster centers are fixed

$\|\cdot\|$ here denotes Euclidean distance

Examples of fuzzy C-means

- We consider the **highest membership** to assign the point to a cluster

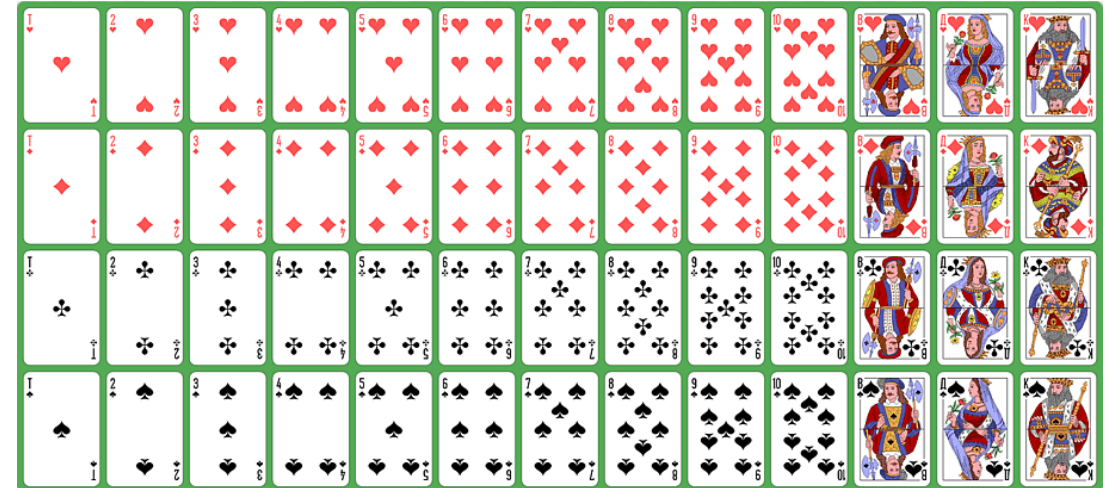


Issues in (fuzzy) clustering

- **Number of clusters**
- **Normalization**
- **Initialization**
- Cluster convexity
- Selection of parameters (e.g., fuzziness ϕ)
- How to deal with categorical variables?
 - Ordinal scale
 - Nominal scale
 - Missing values

Number of clusters

- How many clusters are present in data?
 - Distance dependent, not data dependent
 - Example: how would you cluster a standard 52-card deck of playing cards?
 - According to color: 2 clusters (black, red)
 - According to ranks: 13 clusters (1, 2, ..., 10, jack, queen, and king)
 - According to suits: 4 clusters (♣, ♦, ♥, ♠)
 - And so forth...
- A completely automatic way to determine the exact number of clusters *cannot* exist: **domain expertise** is always necessary (know what you are modeling!)
- Some **heuristics** can help though (e.g., average silhouette, gap statistics), see [Chiang and Mirkin, J Classif 2010] for additional information



Initialization

How to avoid local minima during the optimization? A smart initialization might help

- Randomly select a set of cluster prototypes $\mathbf{V}$
- Randomly select a set of **data points** as cluster centers $\mathbf{V}$
- Randomly initialize the partition matrix $\mathbf{U}$
- Use some information (e.g., cluster center locations) from a **separate clustering step**
- Initialize centers **far away** from data
- Active research area!

FCM and FIS

Remember that a FIS can be built using according to **domain knowledge**

- E.g., universe of discourse of each linguistic variable, fuzzy sets of linguistic terms
- How about building such objects from data?

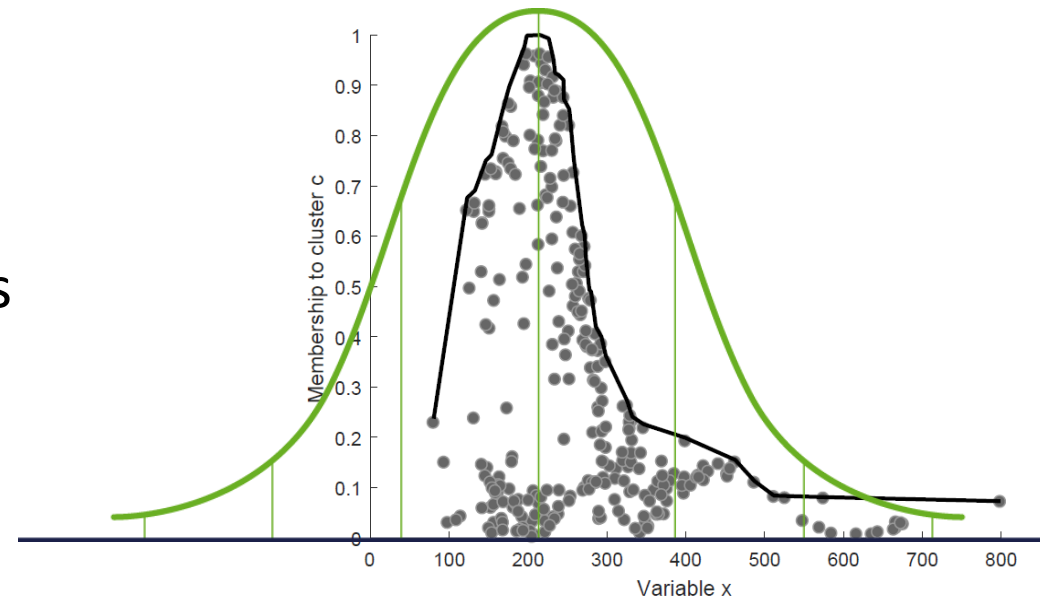
We can use **FCM to cluster** data into groups

- For each group, we can then **project membership values** on linguistic variables
- By looking at the distribution of data, we can **create fuzzy sets** and associate linguistic terms to them
- We can **build rules** by composing such linguistic terms
- Assuming a 0-order Sugeno FIS, we can perform a regression, i.e., **fit the output values** to available data

We created a tool named **pyFUME** that performs the whole process automatically (clustering→rules building→model calibration)

Fitting continuous variables

- For each variable, we can fit a membership function to the convex envelope
- Membership functions should follow data
- We generally leverage Gaussian memberships to maximize **interpretability**



pyFUME main steps (simplified)

intuitive example

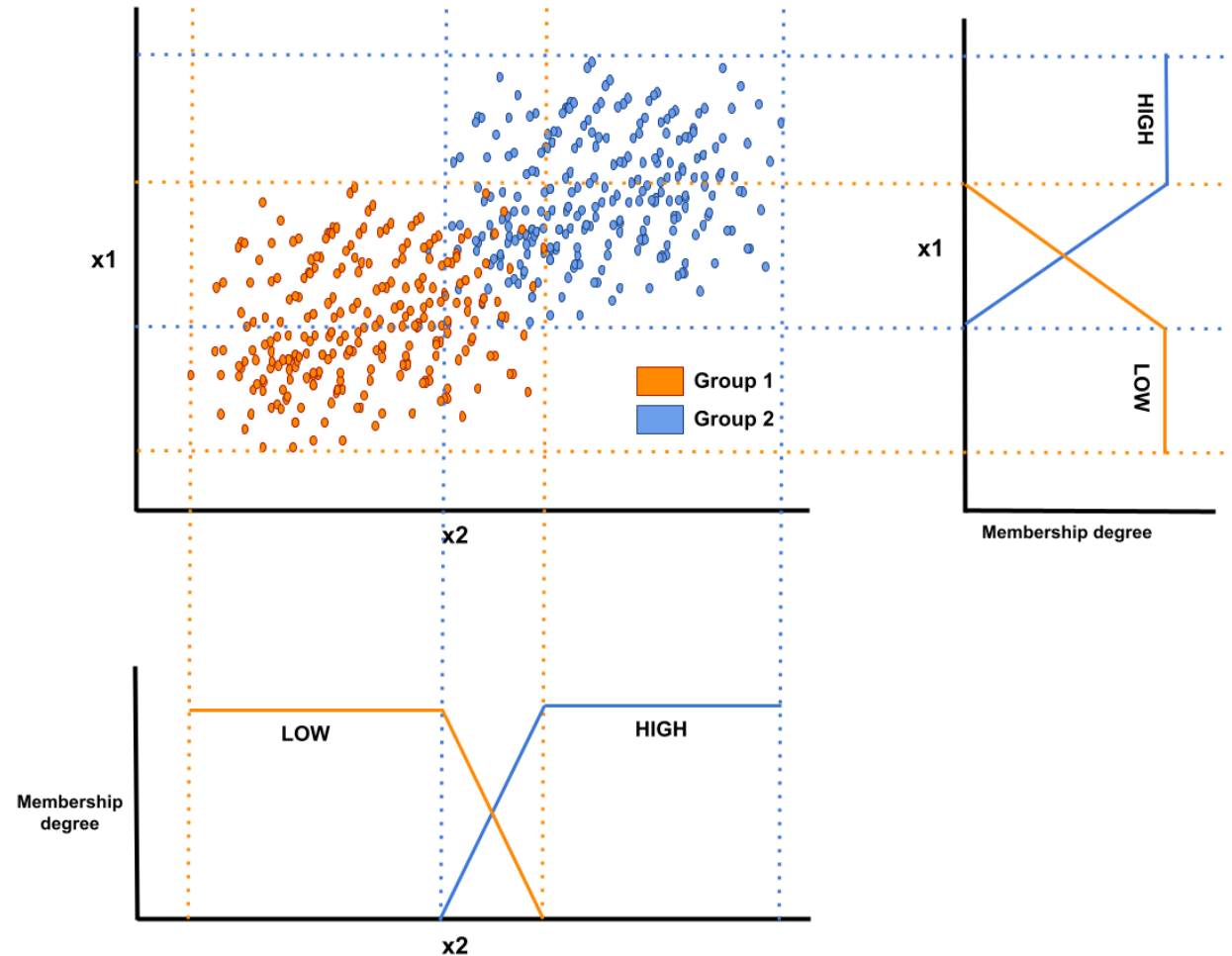
Step 1: clustering

Step 2: mapping clusters to fuzzy sets

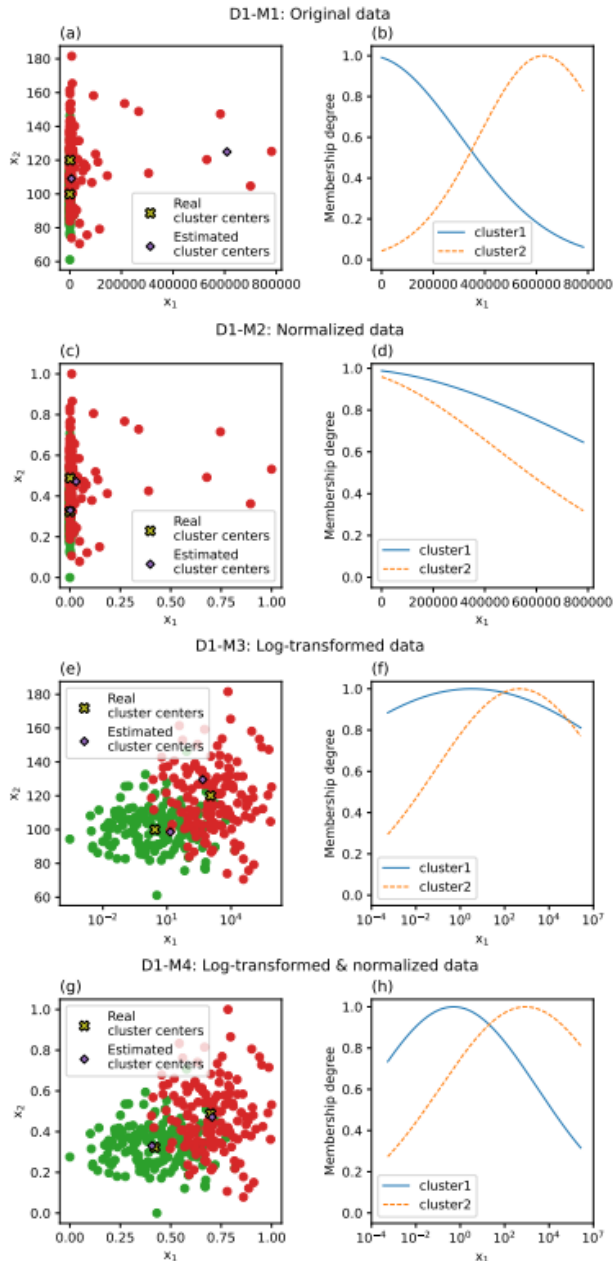
Step 3: rules building

If x_1 is low and x_2 is low, then $y = f(x)$

If x_1 is high and x_2 is high, then $y = f(x)$



pyFUME: evolutionary feature selection, transformation, and cluster estimation



- It is often the case that some variables/features follow a **log-normal distribution**
- That is, they look **normally distributed** when plotted in a **logarithmic scale**
- **pyFUME is fitting Gaussians** in linear space though! → **transformation** of some specific variables is **mandatory** for a proper fitting
- ...and also to maximize interpretability
- Clustering is easier when variables are **normalized in [0,1]**
- **Representation:** not binary but we have 0 (discard variable), 1 (normalize), 2 (normalize and log-transform)
- [Fuchs et al., IEEE CIBCB 2022]

Using pyFUME

- How to install pyFUME: `pip install pyfume`
- How to use pyFUME:

```
from pyfume import pyFUME
FIS = pyFUME(datapath="mydata.csv", nr_clus=3)
model = FIS.get_model()
# you now have a runnable interpretable model for prediction,
# to calculate error, to visualize the fuzzy sets and rules, etc.
```

Summary

- Human reasoning deals with uncertainty and vagueness
- Fuzzy set theory provides a mean to model such uncertainty
- We can create complex models of reality using fuzzy logic and fuzzy rules
- We can perform reasoning on such models using Fuzzy Inference Systems
- Fuzzy C-means clustering allow us to partition data into partially overlapping groups, assisting the creation of complex fuzzy models
- Fuzzy Inference Systems are interpretable universal approximators; they can be tuned to achieve arbitrary precision at the cost of a reduced interpretability
- We can build FIS from data using a machine learning approach (see pyFUME)